Taenia solium

Miguel Perez

Taxonomy

- Domain: Eukaryota
- Kingdom: Animalia
- Phylum: Platyhelminthes
- Subclass: Cestoda
- Order: Cyclophyllidea
- Family: Taeniidae
- Genus: Taenia
- Species: solium

T. solium

Img source: https://www.nikonsmallworld.com/galleries/2017-photomicrography-competition/taenia-solium-everted-scolex

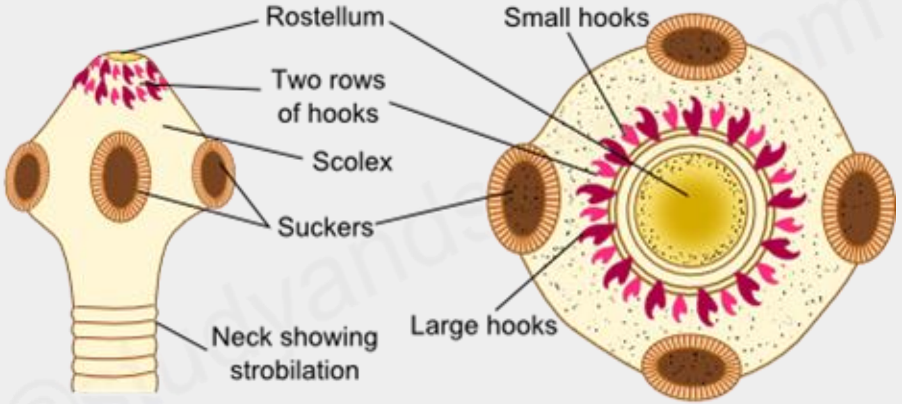
Historical Facts!

- Ancient Disease!
  - Tapeworms in ancient Egyptian mummies [3000-1500 BC]
- Hipólito Unanue [1792]
  - First description of simultaneous taeniasis and cysticercosis

- Friedrich Küchenmeister [19th Century]
  - Demonstration of human intestinal taeniasis via cysticercus
- Kozen Yoshino [1930’s]
  - Confirmed cysticercosis life cycle → tapeworm in humans through group and self experimentation

Morphology

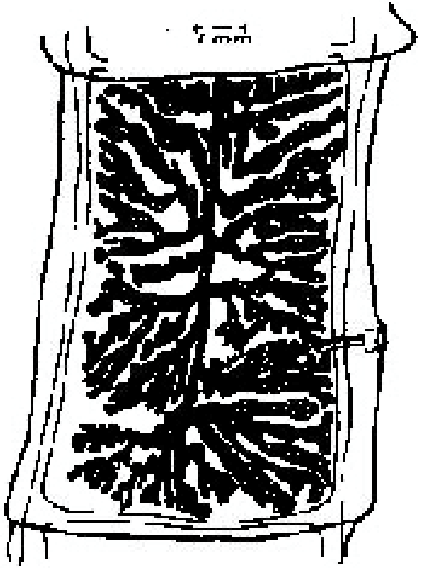
- Mature tapeworm key features:
    - Armed scolex with rostellum
        - 2 circles, 22-32 hooks (alternating)
        - Spheroid scolex



TAENIA SOLIUM - FRONT AND TOP VIEW

- Proglottids
    - Gravid: 7-11 uterine branches
        - Key identifying feature

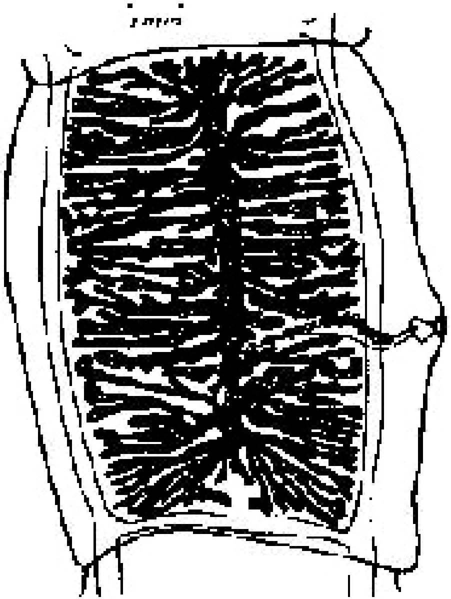
UTERINE BRANCH COMPARISON



GRAVID PROGLOTTID

T. solium

EGG



GRAVID PROGLOTTID

T. saginata

Comparing the two *Taenia* species

Characteristic	*Taenia saginata*	*Taenia solium*
Intermediate Host	Cattle, reindeer	Pig, wild boar
Site of Development	Muscle, viscera	Brain, skin, muscle
Scolex: adult worm	No hooks	Hooks
Scolex: cysticercus	No rostellum	Rostellum & hooks
Proglottids: uterine branches	23 (14 – 32) *	8 (7 –11) *
Passing of proglottids	Single, spontaneous	In groups, passively
Ovary	Two lobes	Three lobes
Vagina: sphincter muscle	Present	Absent

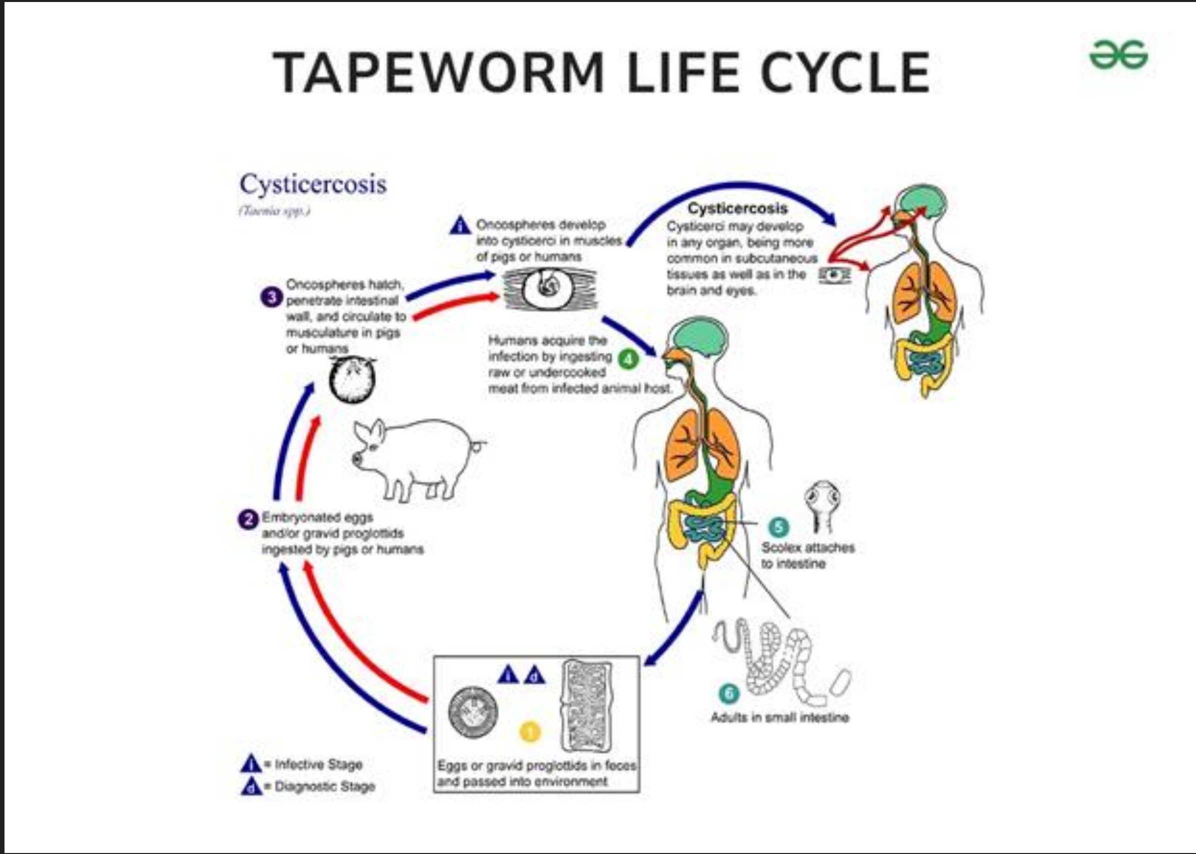
2 Life Cycle Paths:

Solium taeniasis

- **Ingestion of cysticerci**
  - Intermediate host: pig
- Bladder worm attaches to small intestine wall
- Maturation!
  - Within 2-3 months
  - Lifespan: ~25 years
- Gravid proglottids released
  - Groups of 5-6, passively
- Humans are definitive host

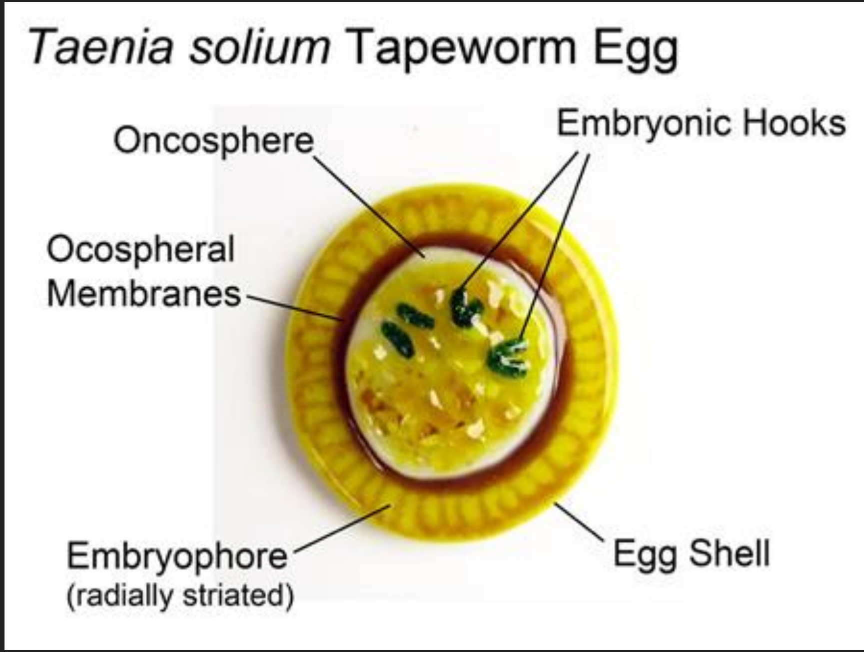
Human cysticercosis

- **Ingestion of eggs/proglottids**
  - Infected human fecal matter
  - Infect others or autoinfection
- Lower intestine → stomach or duodenum
- Travel through blood/lymph
- No tissue/organ is safe
  - Common sites: subcutaneous connective tissues, eyes, brain, muscles, heart, liver, lungs, and coelom.

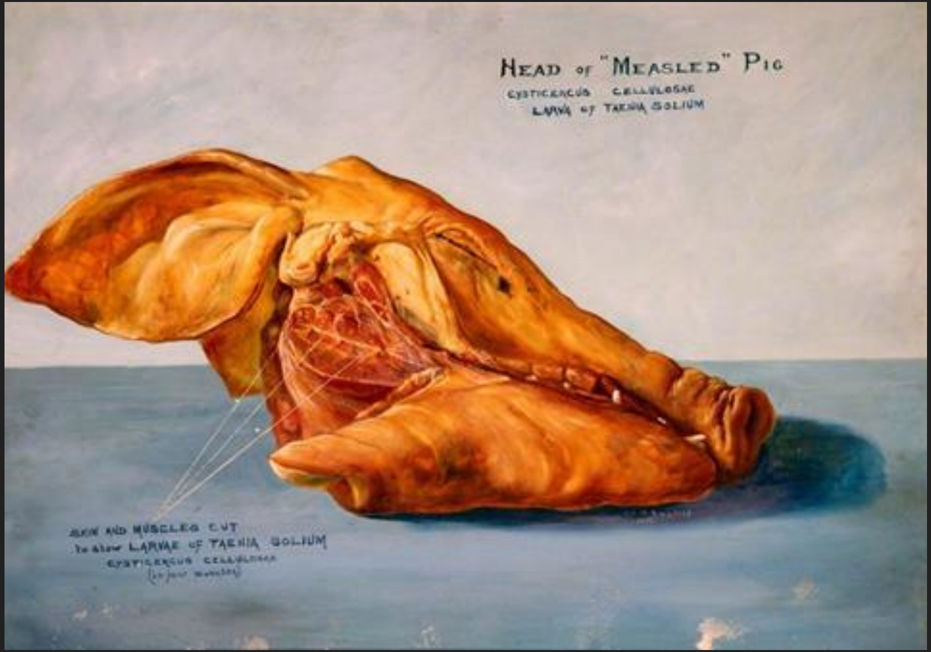


Img source: https://www.geeksforgeeks.org/tapeworm-life-cycle/

- Remember: eat cysticerci → get solium taeniasis; eat eggs/proglottids → get cysticercosis



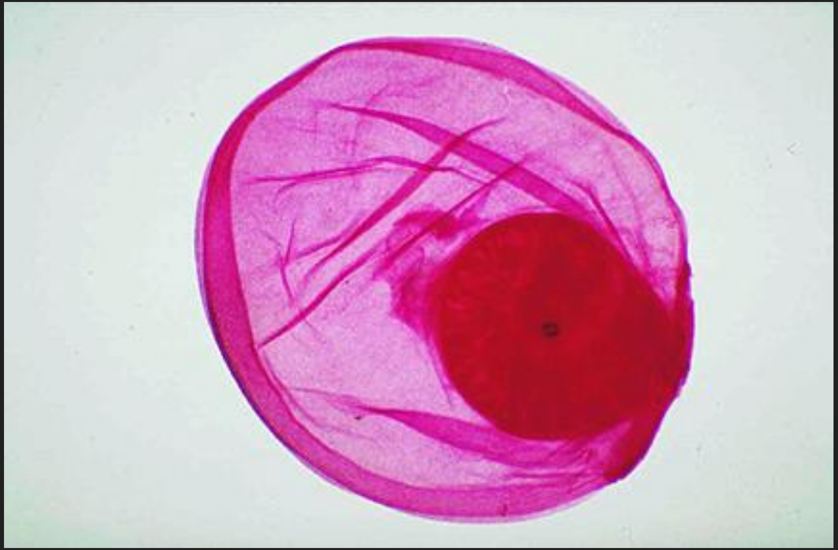
Img source: https://www.deviantart.com/trilobiteglassworks/art/Taenia-solium-Tapeworm-Egg-Fused-Glass-791078227



Img source: https://artuk.org/discover/artworks/head-of-measled-pig-cysticercus-cellulosae-larva-of-tapeworm-taenia-solium-104373

Img source: https://www.cdc.gov/dpdx/taeniasis/index.html

- Note the spheroid scolex



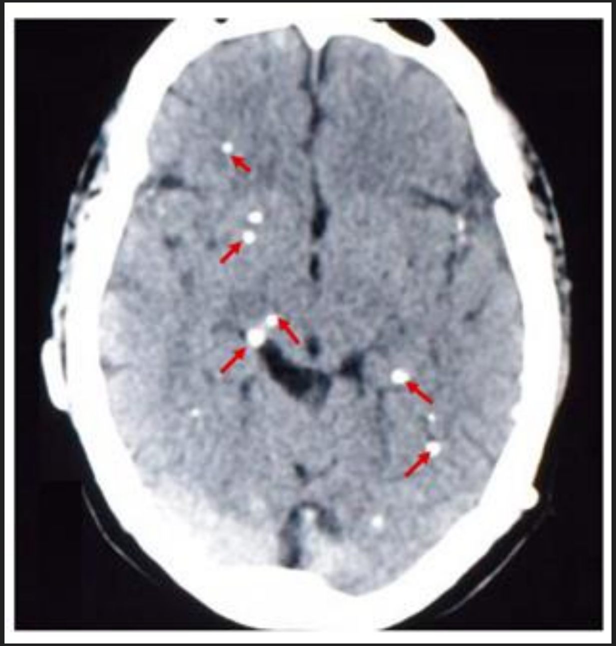
- Fluid-filled cysticercus
  - Note the developing hooks
  - Can be in either pigs or humans

Giant swelling pseudotumor

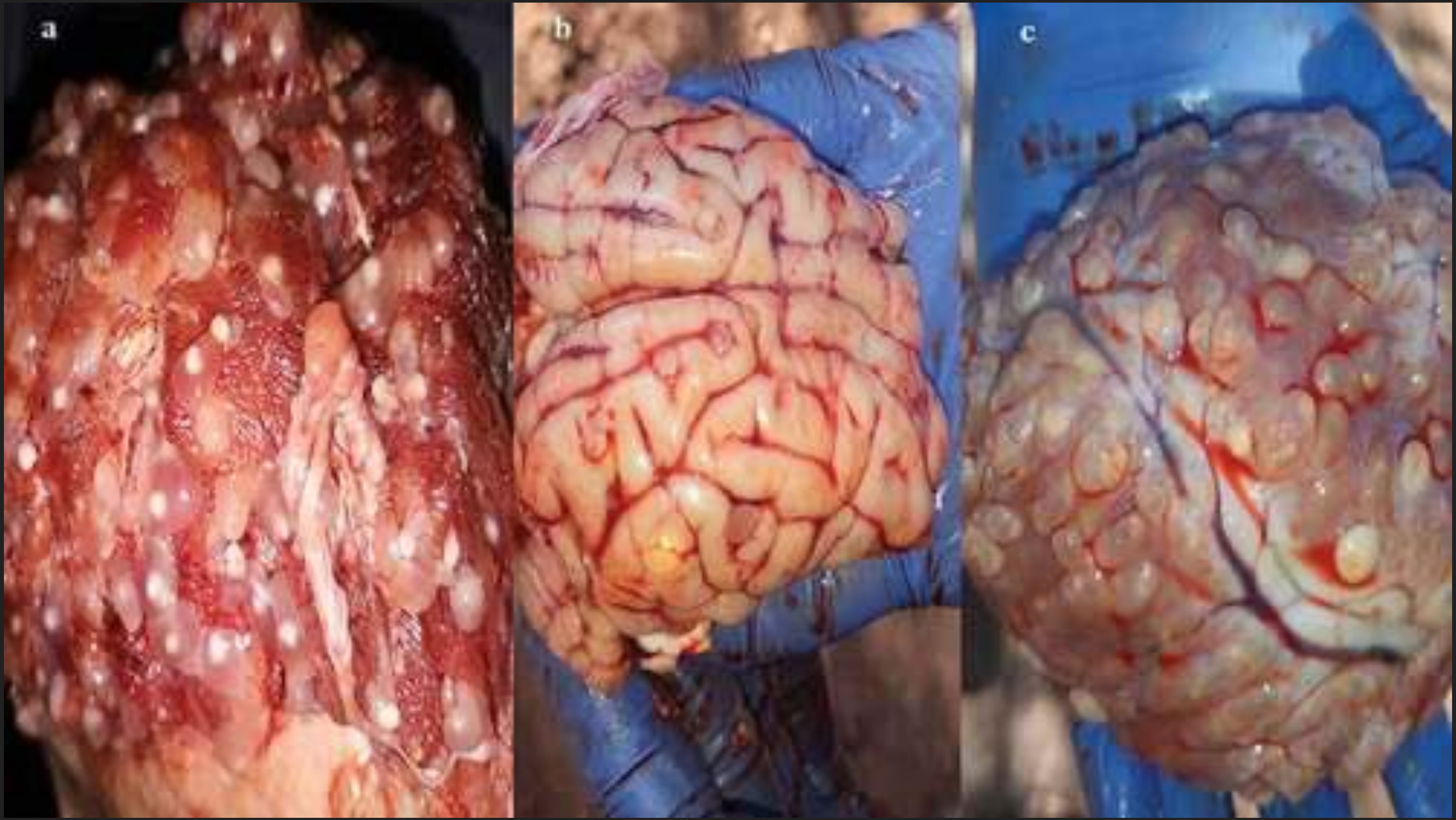
- Fluid build up in cysticercus
  - Can have liters
- Connective tissue forming around cyst

Continuation of human cysticercosis

- Death of cysticerci → calcification
  - Severe inflammatory response
  - 1 year = asymptomatic
  - Many years = serious symptoms
  - Brain infection and calcification
    - Neurocysticercosis
    - Content warning next slide



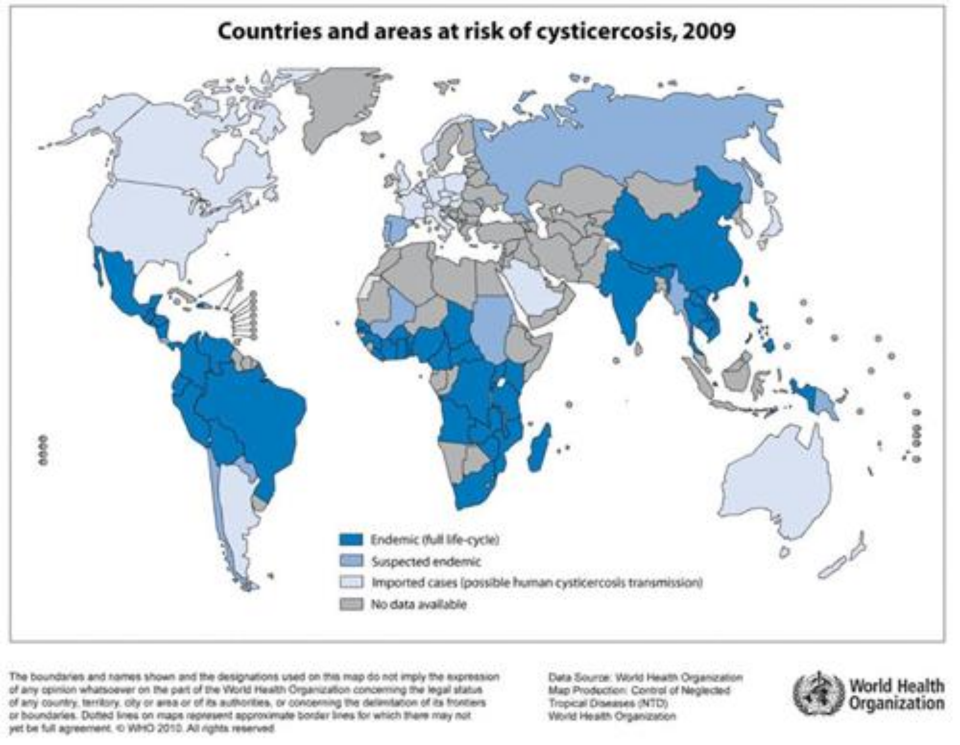
Img source: https://www.mdpi.com/2076-0817/13/1/26



Img source: https://www.tandfonline.com/doi/figure/10.1080/20477724.2022.2083757?scroll=top&needAccess=true

Monsters Inside Me

https://www.youtube.com/watch?v=qITTyuvA5wQ



Img source: https://www.researchgate.net/figure/World-map-showing-the-distribution-of-Taenia-solium-taeniasis-cysticercosis-transmission_fig5_51632511

Epidemiology

Areas of common infection:

- Africa, India, China, South and Central America, Mexico

Impact of culture and diet on infection:

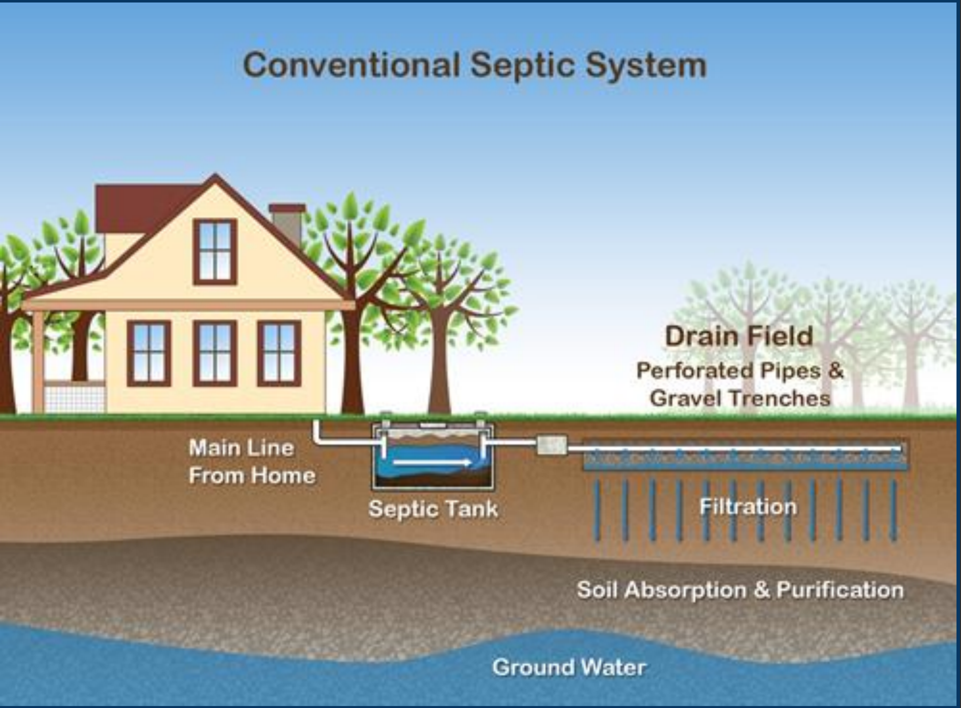
- Muslim and Jewish community
- Vegetarians

Important Aspect

Cysticercosis could happen to anyone

- Vegetarian or non-vegetarian
- Tainted salads or water
- Common in same places as taeniasis

Img source: https://ijdvl.com/extensive-disseminated-cysticercosis/

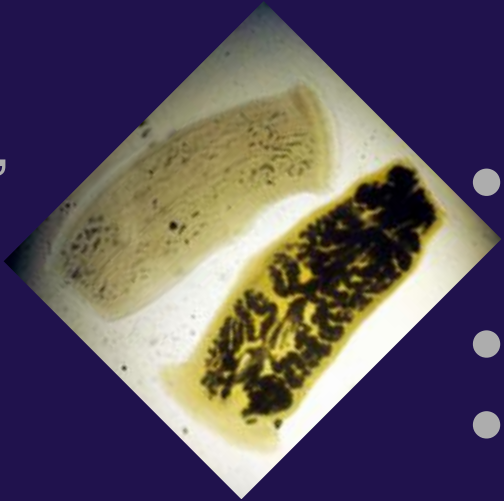


- Common in unsanitary conditions
  - Pigs eating eggs from human feces
  - Flooded pig farms
    - Ex: broken septic tank

Pathogenesis

Solium taeniasis

- Non-specific symptoms
  - Vague abdominal pain
    - Armed scolex
  - Nausea, Diarrhea, Constipation
- Gravid proglottids in stool
  - 5-6 groups,



Img source: https://www.cdc.gov/dpdx/taeniasis/index.html

Cysticercosis

- May remain asymptomatic for many years
- Muscle spasms, weakness, general weakness
- Visible/palpable subcutaneous nodules
- Neurocysticercosis
- Ocular cysticercosis

Treatment, Prevention, and Control

- Praziquantel or Albendazole [1-2 weeks]
  - For both solium taeniasis and cysticercosis
  - 5-10 mg/kg Praziquantel
  - 400 mg Albendazole
- Surgery for cysticerci

- Meat inspection
- Properly cooking pork
- Safe disposal of human waste
- Proper washing of vegetables
- Sexual habits and proper hygiene
- Boiling and filtering of water
- Raw fruits and vegetables in endemic regions

Case Study 1 - Neurocysticercosis

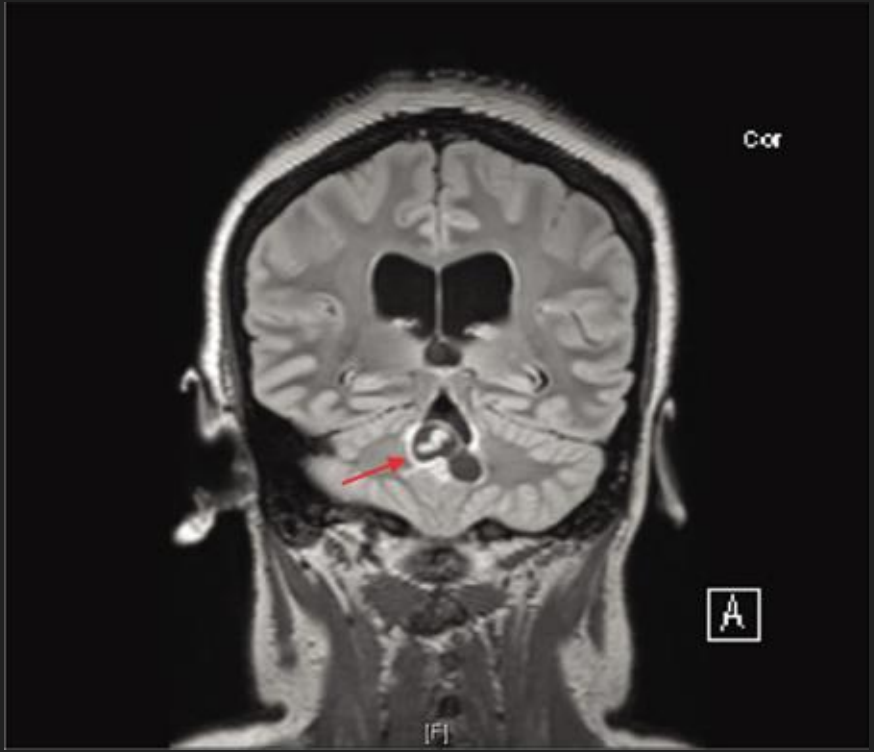
- 36-year old female
- History of adult onset seizures
- Born and raised in Mexico
  - Moved to US at age 8
- First seizure: 10 years prior
- Denied report of any intracranial lesions

Further examination

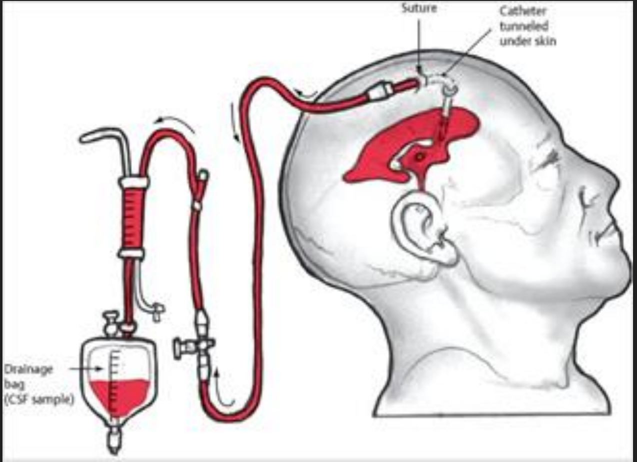
- Worsening headaches
  - Worse at night and in the morning, but better throughout day
- CT scan
  - Ventriculomegaly
    - Cerebrospinal fluid (CSF) buildup in brain ventricles
  - Transependymal flow
    - CSF in ventricles → brain tissue
  - Obstruction at fourth ventricle level
- No neurological deficits found

Further examination pt. 2

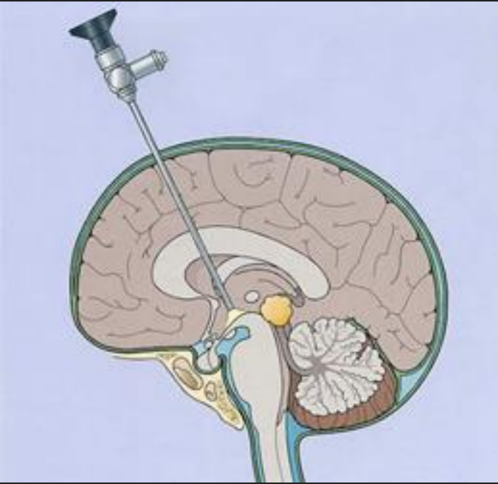
- MRI
  - Lobulated cystic mass within fourth ventricle
  - Internal linear enhancement
    - Increased brightness in area
    - Indicative of neurocysticercosis
  - Worm within cyst



- Elective surgery scheduled
- Right frontal burr hole (hole in skull)
  - External Ventricular Drain (EVD) for CSF draining
  - Endoscopic Third Ventriculostomy (ETV) = bypass for CSF to travel through to normal pathway

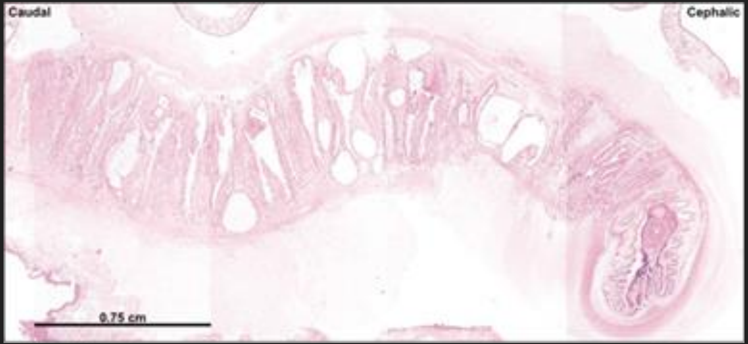


Img source: https://neupsykey.com/external-ventricular-drain/



Img source: http://oxfordneurosurgeon.com/treatments/endoscopic-surgery_intraventricular-endoscopy

- Suboccipital craniotomy
  - Removal of skull at base of the head
  - Access to back of the brain
- Removal of cystic mass *en bloc* (in one piece)



- Identification of mature *T. solium* plerocercoid larvae (1.2 X 0.5 X 2 cm)

Post Op

- No neurological deficits
- EVD removed on day 2
- Ophthalmologic exam
  - Concern for chronic, inactive choroidal inflammatory process
- Suspected of auto-infection
  - History of seizures
  - Recent presentation with active lesion

- Albendazole treatment (21 days)
- Dexamethasone taper (14 days)
  - Guard against aseptic meningitis
- Single dose of Praziquantel
  - Gastrointestinal auto-infection risk reduction

Case Study 2: Biliary solium taeniasis

- 70-year old Korean man
- Recurrent vomiting and colicky pain
- Fever (38.0°C), physical exam conducted
- Showed positivity for Murphy’s sign
    - Gallbladder inflammation

?

- Denied history of:
    - Hepatobiliary disease (bile ducts, liver, gallbladder)
    - Parasitic infections
    - Digestive problems
    - Passing of parasites in feces
    - Eating uncooked food

Inflammatory response

Severe inflammatory status

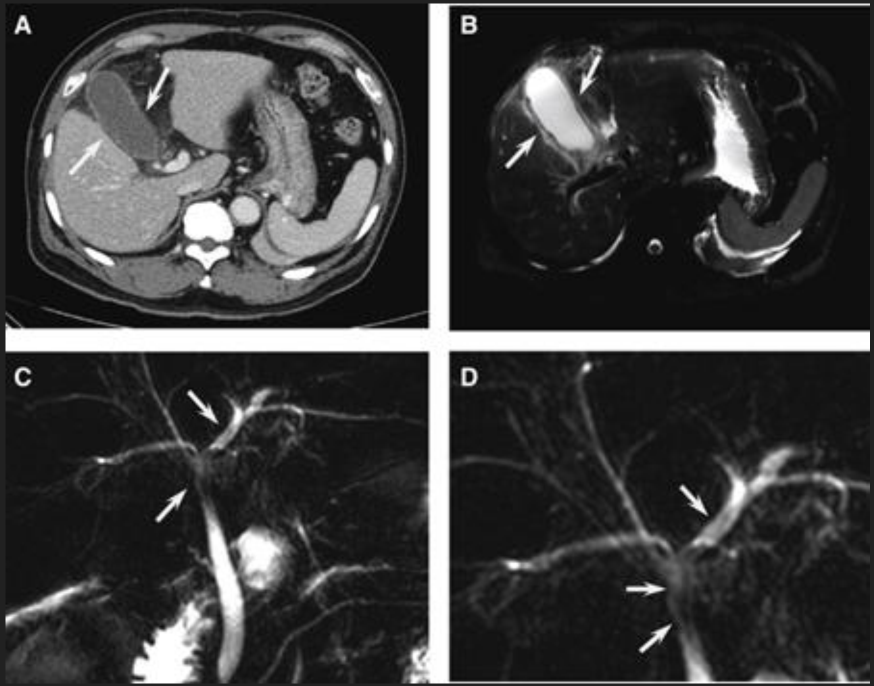
	Patient Readings	Normal Readings
WBC count	20.6 * 10^9/L	4.5 - 11.0 * 10^9/L
LDH serum levels	275 IU/L	125 - 220 IU/L
CRP serum levels	26.84 mg/dL	0.3 - 1.0 mg/dL
Fasting glucose serum levels	169 mg/dL	70 - 100 mg/dL
HbA1c range	7.60%	6.50%

Diabetes mellitus

- Hyperglycemia

CT Scan

- Thickened gallbladder wall
  - Suggests tissue death of wall due to reduced blood flow
  - Tissue swelling around gallbladder → inflammation



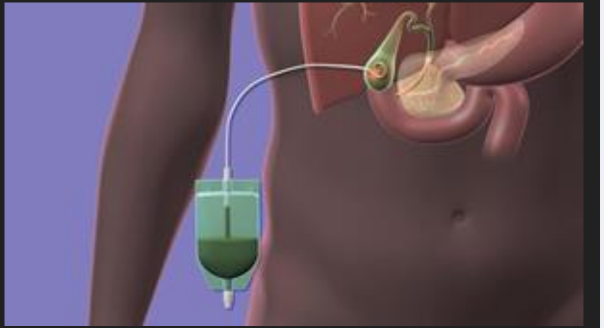
MRCP Image

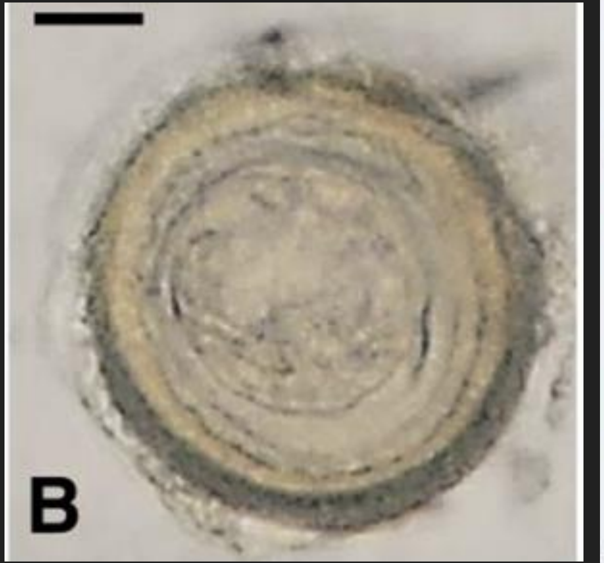
- Darkened linear structures within intra- and extrahepatic bile ducts

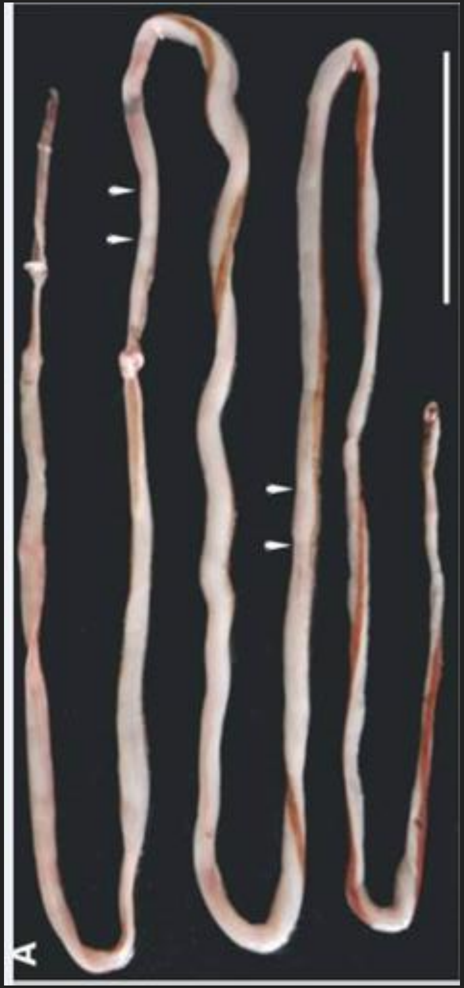
- Immediately prescribed ceftriaxone and metronidazole

3rd day of admission

- Percutaneous transhepatic gallbladder drainage
- During fluid irrigation of catheter, long-bodied parasite was drawn out
- Highly degenerated proglottids were found on posterior end
- Striated embryophore with hooklets found

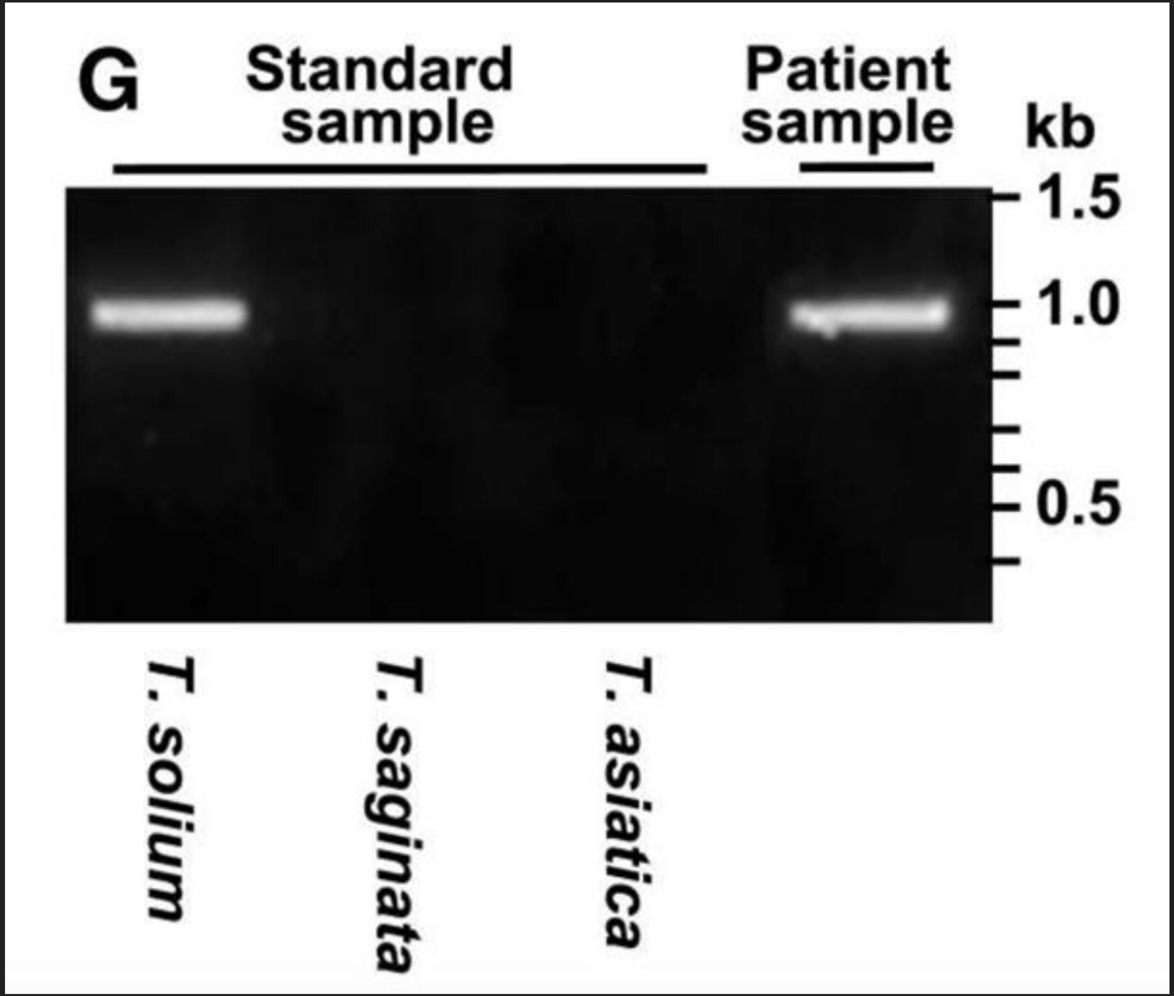






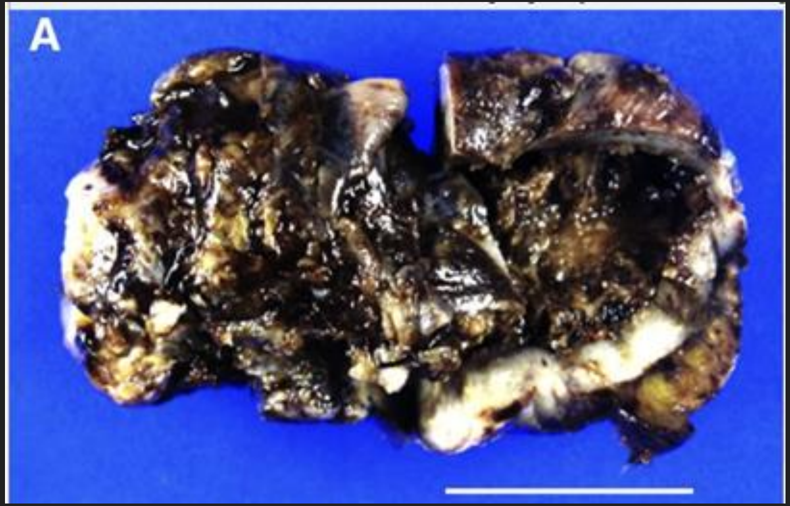
ID-ing the parasite

- PCR test conducted with worm DNA



Post Worm Removal

- Worm removal → gastrointestinal symptoms subsided
- Treated with single dose of Praziquantel (10 mg/kg)
- Persistent fever with inflammatory signs → inflamed gallbladder removal
  - Laparoscopic cholecystectomy
- Residual parasitic eggs/worms not observed
- Recovery after 1 month
  - All serum levels returned to normal

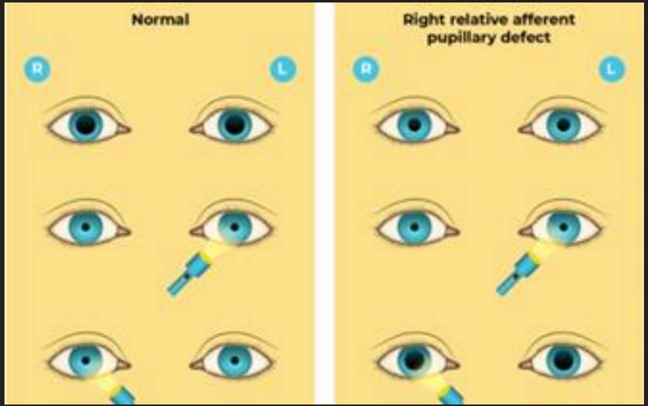


Case Study 3 - Subretinal Cysticercosis

- 42-year old female
- Decrease in right eye vision
  - 1 month
  - Painless, progressive
  - Associated with floaters

- 10 years without vision in left eye
- History of left-sided hemiplegia and Multiple Sclerosis for 10 years
- No history of Diabetes Mellitus or Hypertension
- Vegetarian diet for 2 years, ate pork meat 2 years prior

Eye tests and observations

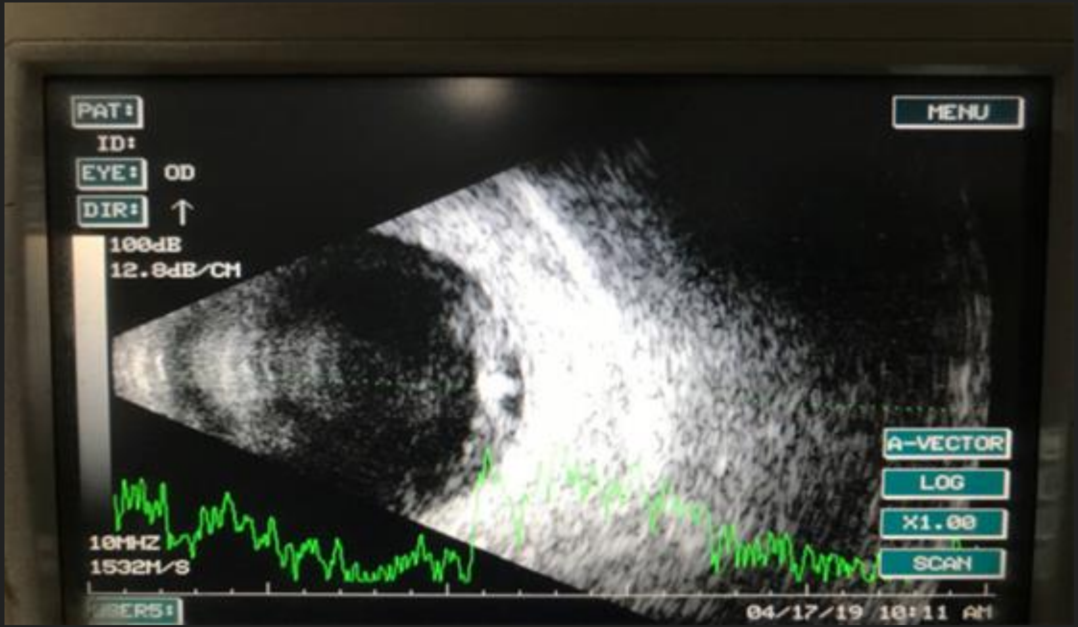


- Best Corrected Visual Acuity (BCVA)
  - Right eye: 5/60
  - Left eye: light perception only
- Intraocular pressure levels were normal
  - Typical range of 10-20 mmHg
  - 11 mmHg for the right, 15 mmHg for the left
- Rapid Afferent Pupillary Defect in left eye
  - Irregular response to light stimuli

- Whitish round cystic mass found
  - Posterior vitreous attachment
  - Vitreous = referring to the jelly-like substance inside the eyeball

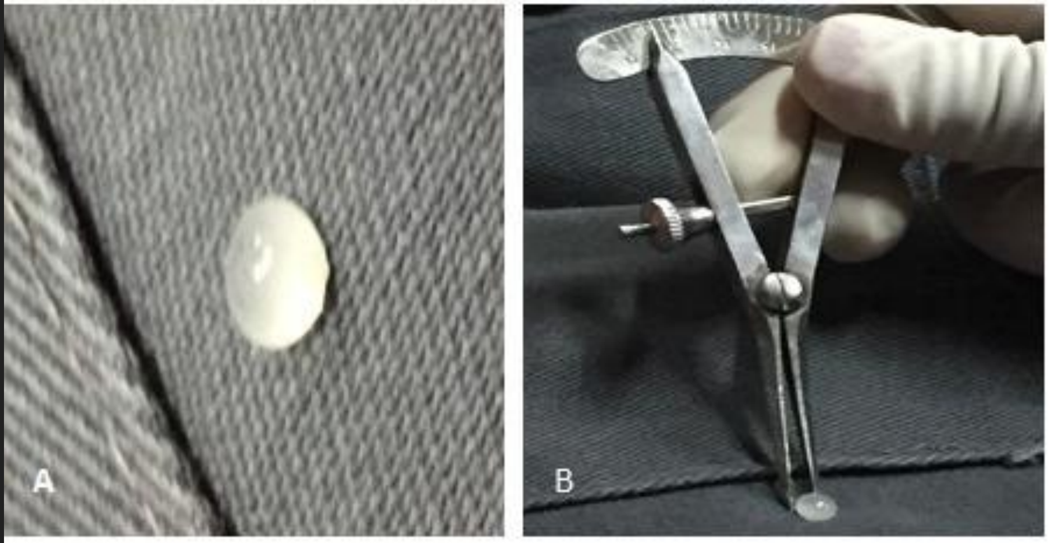
Further examination of the cyst

- Dense white material on cyst
  - Cysticercoid with scolex
- Brightness scan
  - Highly reflective elevation on the dome suggests scolex



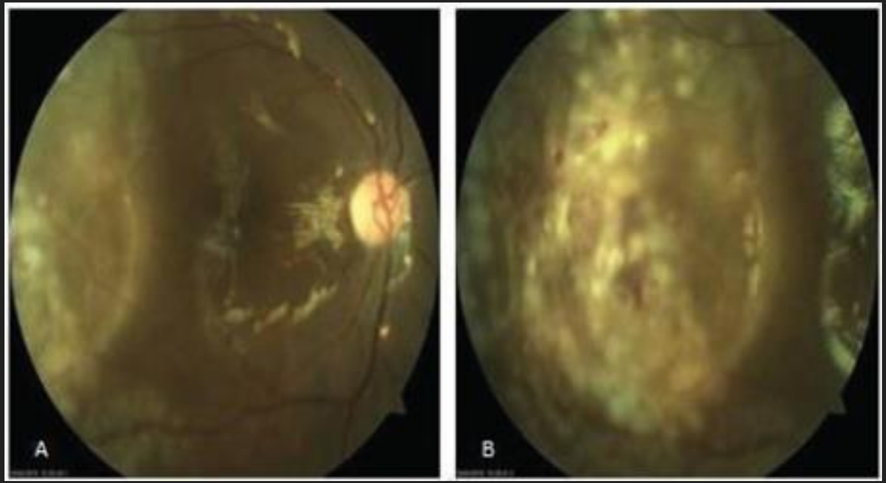
- MRI
  - Lesions on the brain
    - Chronic hemorrhagic lesions
    - Represents calcific foci (specific area of abnormality)
  - Possible calcified cysticerci
- Blood tests and serological tests were normal
- No parasites found in stool
- Provisional diagnosis:
  - Right eye subretinal cyst due to cysticercus
  - Left eye atrophy due to multiple sclerosis

Operation



- Pars plana vitrectomy of right eye
  - Incision in eye to access vitreous gel
- Subretinal cyst removed
  - Sent out for histopathological examination
- Gross examination
  - 5 mm cyst
  - Whitish material within cyst suggests cysticerci with scolex

- BCVA on first postoperative day
  - 1/60 for right eye
  - Intraocular pressure at 10 mmHg
- Fundus examination
  - Back of the eye
  - Well-attached retina
  - Laser marks
  - Silicone oil
    - Fill space of vitreous in post op

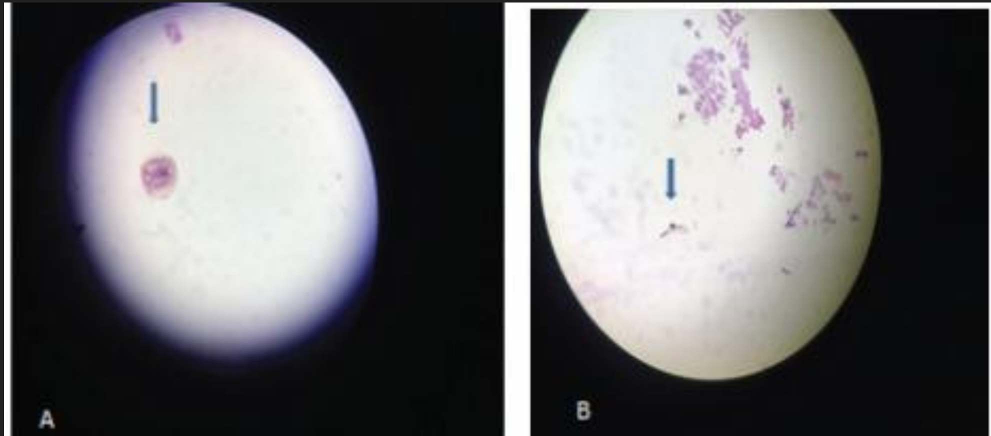


2 weeks later

- BCVA of 6/36 with +6.00 Dioptre spherical lens
  - To correct sight
- Fundus examination similar to before

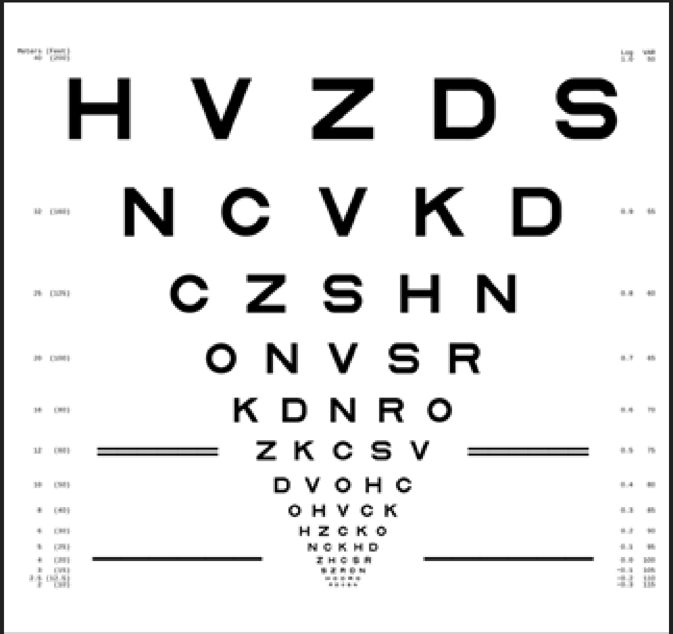
Histopathology

- Presence of suckers and hooks of scolex
  - Diagnosis of cysticercosis ✓



  - Sucker on A, scolex on B

- Patient prescribed glasses
  - 6/36 BCVA
  - Over the course of 1 month



3 months later

- Silicone oil removed
- Final BCVA: 6/12

Question 1:

What was the key indicator of neurocysticercosis in case study 1?

MRI scan of the cyst with a worm in it

Question 2

Why was the taeniasis in case study 2 particularly bizarre?

Because the worm was found in the bile duct, not the small intestine

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